Ch. 14 CW #1

Principles of Microeconomics

November 30, 2025

Instructions

Please answer the following on a separate sheet of paper and turn them in on Gradescope by next class.

Problems

1. How and why does a firm's average-total-cost curve in the short run differ from its average-total-cost curve in the long run?

2. Define economies of scale and explain why they might arise. Define diseconomies of scale and explain why they might arise.

3. Consider the following cost information for a pizzeria:

Quantity	**Total Cost**	**Variable Cost**
0 dozen pizzas	\$300	\$0
1	350	50
2	390	90
3	420	120
4	450	150
5	490	190
6	540	240

   (a) What is the pizzeria's fixed cost?

   (b) Construct a table in which you calculate the marginal cost per dozen pizzas using the information on total cost. Also, calculate the marginal cost per dozen pizzas using the information on variable cost. What is the relationship between these sets of numbers? Explain.

4. You cousin Vinnie owns a painting company with fixed costs of \$200 and the following schedule for variable costs:

Quantity of Houses Painted per Month	1	2	3	4	5	6	7
Variable Costs	\$10	\$20	\$40	\$80	\$160	\$320	\$640

Calculate average fixed cost, average variable cost, and average total cost for each quantity. What is the efficient scale of the painting company?

5. The city government is considering two tax proposals:

   - A lump-sum tax of \$300 on each producer of hamburgers
   - A tax of \$1 per burger, paid by producers of hamburgers

   (a) Which of the following curves – average fixed cost, average variable cost, average total cost, and marginal cost – would shift as a result of the lump-sum tax? Why? Show this in a graph. Label the graph as precisely as possible.

   (b) Which of these same four curves would shift as a result of the per-burger tax? Why? Show this in a new graph. Label the graph as precisely as possible.

6. Jane's Juice Bar has the following cost schedules:

Quantity	**Variable Cost**	**Total Cost**
0 vats of juice	\$0	\$30
1	10	40
2	25	55
3	45	75
4	70	100
5	100	130
6	135	165

   (a) Calculate average variable cost, average total cost, and marginal cost for each quantity.

   (b) Graph all three curves. What is the relationship between the marginal-cost curve and the average-total-cost curve? Between the marginal-cost curve and the average-variable-cost curve? Explain.

7. Consider the following table of long-run total costs for three different firms:

Quantity	1	2	3	4	5	6	7
Firm A	\$60	\$70	\$80	\$90	\$100	\$110	\$120
Firm B	11	24	39	56	75	96	119
Firm C	21	34	49	66	85	106	129

Does each of these firms experience economies of scale or diseconomies of scale?